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Testing and performance analysis of an integrated electromagnetic and hydraulic retarder for heavy-duty vehicles☆

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ABSTRACT

Hydraulic retarder and eddy current retarder are important auxiliary braking component, which installed on heavy-duty vehicles. However, the hydraulic retarder has poor braking performance at low speeds, while the eddy-current retarder presents serious thermal recession at high speed. To overcome these issues and ensure safe driving during downhill conditions, a novel Integrated Electromagnetic and Hydraulic Retarder (IEHR) is proposed. The design includes a rotor with blades, double salient poles, and a coil built in a C-type stator. The electromagnetic field model of the IEHR is established and analyzed using the Finite Element Method. Moreover, the nonlinear eddy viscosity model of the flow field is generated and analyzed using the Computational Fluid Dynamics. The Blend braking control strategy is adopted to distribute the braking torque at high speed and low speed The IEHR performance was tested at a test bench. The test results show that the calculated braking characteristics of the electromagnetic and hydraulic subsystems were reasonable agreement with the measured results. The road test was carried out on a vehicle with a total mass of 49 tons. The deceleration and torque peaked at 0.75 m/s^2 and 3620 N m, respectively. These results indicated that the IEHR has better performance than conventional hydraulic retarders and eddy current retarder, and is more suitable for heavy vehicles.

1. Introduction

Eddy current retarders (ECBs) and hydraulic retarders (HRs) are commonly used to slowdown heavy vehicles (Zheng et al., 2016; Ye et al., 2019a), during non-emergency conditions. They can effectively reduce the required braking strength and frequency of operation of the traditional mechanical braking systems (Zheng et al., 2017). Heavy-duty vehicles need to have great torque to consume the vehicle's gravitational potential energy when they go downhill at a constant speed. Therefore, retarders for these kinds of vehicles are required to have large braking torque, particularly in the full-speed regime.

ECBs are used to convert the kinetic energy of the vehicle into heat at the rotor. ECBs are mainly used on buses and have many advantages like a simple structure, fast response speed and good braking performance at a low vehicle speed. Liu et al. proposed the structure of an eddy current retarder and measured its brake characteristics, temperature characteristics and power consumption with the test bed (Liu et al., 2011). The structure of their ECB and its braking torque-speed curve are shown in Fig. 1. The braking torque rapidly decreases with breaking time, approximately 40%, due to the increased temperatures at the rotor (Zhang et al., 2015). The brake torque of these ECBs reached its maximum value at a speed of approximately 500–700 rpm.

In order to obtain larger braking torques and improve the serious thermal recession of ECBs, researchers have developed a variety of water-cooled eddy-current retarders (Li et al., 2021; Tian et al., 2020; G et al., 2019). Ye et al. proposed a dual air-gap and liquid-cooled, eddy-current retarder that improves heat fade of the continuous braking and the large rotor weight of traditional ECBs for heavy vehicles (Ye et al., 2019b). An ECBs and heating system has also been proposed and designed to increase the low recovery efficiency of regenerative braking energy (Ye et al., 2019c). Although the braking torque of water-cooled ECBs can meet typical vehicle demands, its quality is much higher than that of the hydraulic retarder (Gao et al., 2020). The ECBs have the advantages of large braking torque and fast response speed, however, the ECBs have not been widely used in heavy trucks, mostly due to their large mass and serious heat recession.

On the other hand, HRs are used to convert the vehicle kinetic energy into internal energy and heat of the fluid. HRs are mainly used in

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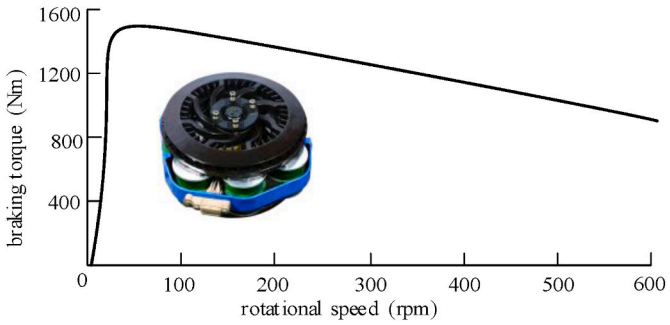


Fig. 1. Brake torque-speed curve of an eddy current retarder.

medium and heavy-duty commercial vehicles to ensure safe driving (Wang et al., 2021; Liu et al., 2018). They have excellent braking performance at high speed, small volume and light weight (Li et al., 2014). There are many factors affecting the braking torque of the HRs, including the liquid filling capacity, the shape and effective diameter of the circulatory circles and the number and inclination mode and angle of the blades (Chen et al., 2016). Research on HRs has been gradually developed in the past years to obtain large braking torques (Wei et al., 2019a). The different rotor blade lean angles on HRs affects their braking performance. Yan (Yan et al., 2015) used the uniform density and gas-liquid stratification methods to derive the HR braking torque formula under different working and filling conditions. They also used the two-phase flow model to conduct numerical studies of the internal flow field. Nevertheless, when the vehicle speed is lower than 20 km/h and the gradient of the road is 6%–7%, the hydraulic retarder is not often used driving downhill. The HRs have poor braking performance at low speeds.

Based on the above analysis, the ECBs has good braking performance at low speed, while the HRs has good braking performance at high speed, we propose an Integrated Electromagnetic and Hydraulic Retarder structure. We developed and analyzed the electromagnetic field using the finite element method. Moreover, the nonlinear eddy viscosity model of the flow field was analyzed using the Computational fluid dynamics (CFD). In addition, the braking performance were tested at a test bench.

2. Structure and working principle

The IEHR is shown in Fig. 2. The design includes a rotor with blades, double salient poles, and a coil built in a C-type stator. The inner and outer sides of the rotor are composed of salient poles, and one side of the rotor has blades. The structure of traditional HRs is complicated, with the blades adopting a strongly inclined structure. On the other hand, all the blades of the IEHR have an axial straight shape, which helps accelerate and steer the working fluid.

When the rotor rotates together with the drive shaft of the rear axis differential at a low speed, the excitation current is provided by the storage battery. A magnetic field appears around the coil and eddy currents generate an alternating magnetic field on the inner surface of the C-type stator, opposite to each magnetic pole. The braking torque is then produced that slows down the vehicle. A large amount of heat is generated on the C-type stator when the coil is energized. The heat is taken away from the circulating liquid coolant at the water tank.

When the rotor rotates together with the drive shaft at high speed, the solenoid valve is opened by the control module. The fluid enters the hydraulic circulatory circle from the water tank. The working chamber is filled with liquid, and the rotor acts as a power source driving the fluid to circulate in a vortex flow. The braking torque is continuously generated owing to the fluid constantly moving in the chamber by the rotating rotor. The working fluid flows out of the chamber and then to the water tank. Eventually, the IEHR converts the kinetic energy of the vehicle into the thermal energy of the working fluid. The heat is taken away by the water tank and a cooling fan.

3. Design and modeling

The braking characteristics of the IEHR are derived from both the electromagnetic field model and flow field model. The basic structure of the proposed IEHR is shown in Fig. 3. Although the structure of the

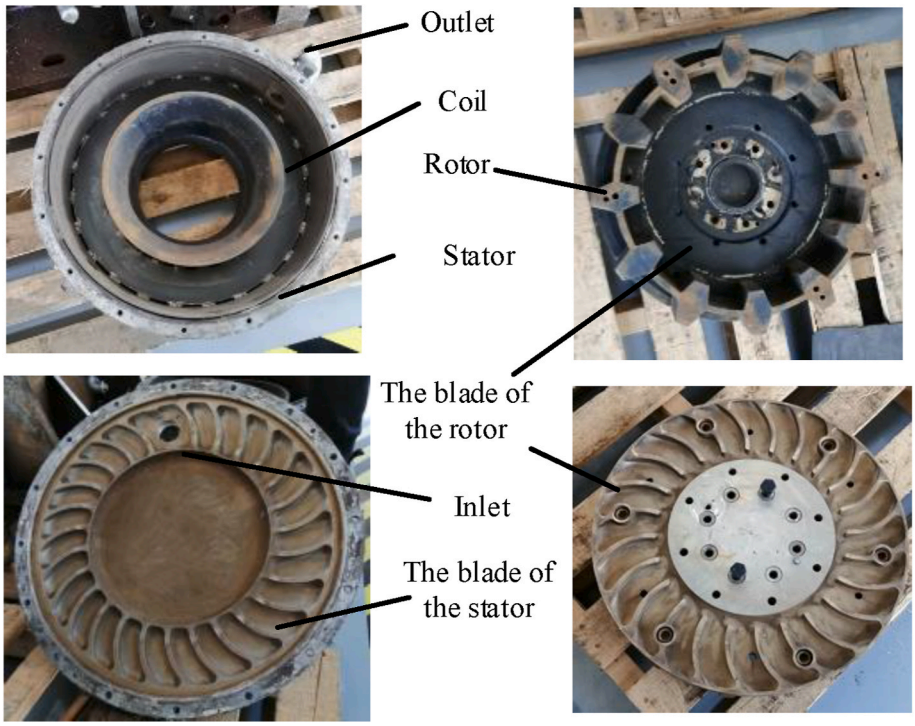


Fig. 2. Structure of the IEHR.

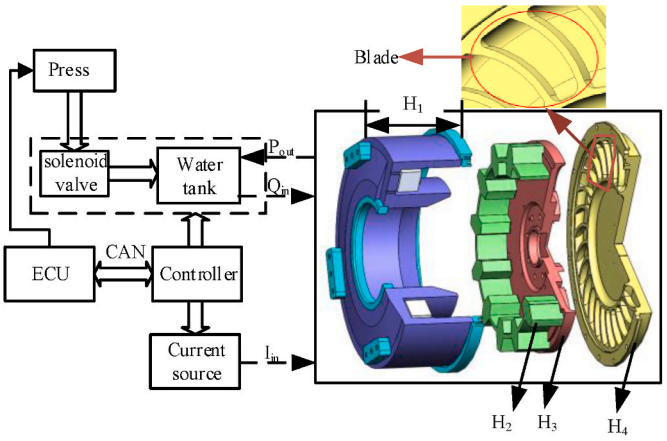


Fig. 3. Schematic diagram of the IEHR.

hydraulic and electromagnetic subsystems is relatively independent of each other, they do relate via the coolant. The control module of IEHR shares data with the electronic control unit (ECU) of the vehicle, including vehicle speed, ABS, braking signals, etc. The IEHR had one nonrotating excitation coil which is located in the C-type stator, and is excited by the current from a storage battery via the rectifier. The control system supplies the coil current.

In order to calculate the braking torque of the IEHR, the structural parameters are determined firstly according to the target braking torque. The flow chart of the IEHR performance analysis is shown in Fig. 4. The main structural parameters include the shape and number of salient poles, the thickness of the stator, the shape and number of blades, the materials of each part, etc. The electromagnetic field and flow field are established. The static magnetic field analysis was carried out to obtain the relationship between magnetomotive force and air gap average flux density. The braking power and skin depth are assumed as the boundary conditions for the transient conjugate heat transfer analysis of the stator. The working principle of the IEHR is to convert the kinetic energy of the vehicle into heat energy, which is taken away by the circulating coolant. In order to obtain the average temperature of the stator heat source area, the transient heat transfer analysis needs to be carried out for a certain period of time. The better the heat dissipation performance of the IEHR, the smaller the heat recession during continuous braking. If the analysis result reaches the target braking torque, the calculation shall be stopped. If the analysis results show that the target braking torque cannot be reached, the model parameters need to be modified and recalculate.

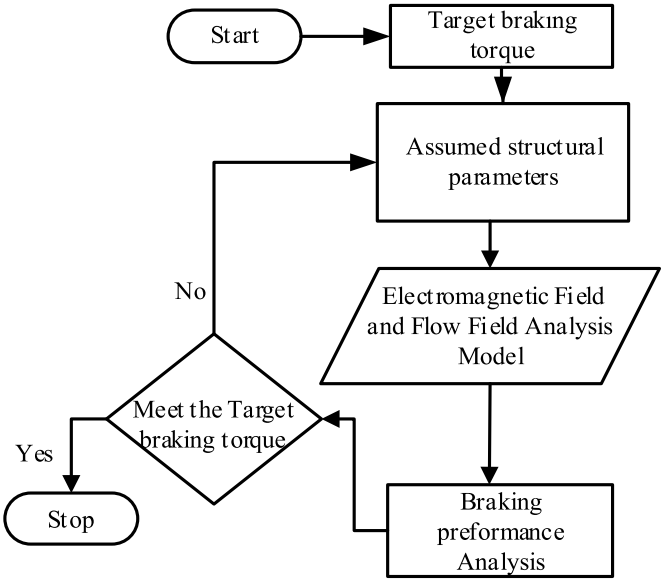


Fig. 4. Flow chart of the IEHR performance analysis.

3.1. Finite-element electromagnetic field model

The electromagnetic field analysis includes the calculation of the magnetic circuit and the numerical analysis of electromagnetic field model with the J-MAG software (Li et al., 2021; Tian et al., 2021).

3.1.1. Calculation of the magnetic circuit

The magnetic circuit of the IEHR is shown in Fig. 5. An air gap is formed between the rotor and the C-type stator. The inner surface of the C-type stator generates an alternating magnetic field at each salient pole of the rotor. This, in turn, establishes eddy currents on the surface of the C-type stator. We use an equivalent magnetic circuit to estimate the double air gap flux density.

According to Kirchhoff's loop rule for magnetic circuits, the total reluctance (R_i) consists of the reluctance of the rotor, the double air gap and the stator. The relevant equations are:

$$F = NI \tag{1}$$

$$R_i = \frac{l_i}{\mu_i S_i} \ (i = \mathrm{ab}, \mathrm{bc}, \ldots, \mathrm{ja}) \tag{2}$$

where F is the magnetic motive force, N is the number of turns of the coil, I is the current of the coil, R_i (i = ab, bc, …, ja) are the magnetic reluctances of the magnetic heads of rotor and stator, respectively, l_i is the flux path length, μ_i is the magnetic permeability, and S_i is the flux area.

The static-state flux density of the double air gap can then be obtained from:

$$B_{\mathrm{ab}} = \frac{\mu_0 \left(NI - \sum_{i=bc}^{ja} \Phi_i R_i \right)}{l_{ab}} \tag{3}$$

3.1.2. Numerical analysis

The basic structure considered in our finite element model of the electromagnetic field is shown in Fig. 6. The main components are the C-type stator, one nonrotating excitation coil, the air gap and the rotor with salient poles. The resulting equations of the A–φ method are (Jiao et al., 2014; Zhang et al., 2014):

$$\nabla \times (\nu \nabla \times A) = J_S - \sigma \left(\frac{\partial A}{\partial t} + \nabla \varphi \right) \tag{4}$$

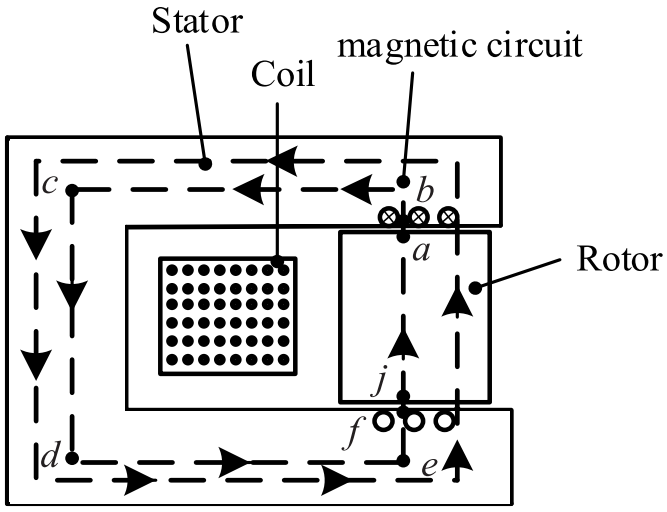


Fig. 5. The simplified magnetic circuit of the braking system.

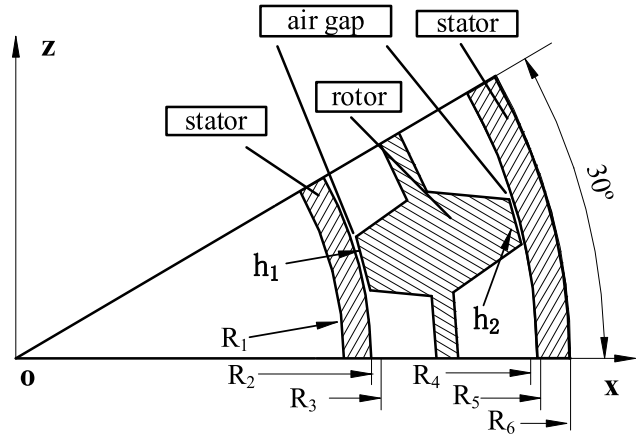


Fig. 6. The analysis model of the electromagnetic field.

$$\nabla \cdot \left[-\sigma\left(\frac{\partial A}{\partial t} + \nabla\varphi\right)\right] = 0 \tag{5}$$

where A is the magnetic vector potential, φ is the electric scalar potential, J_s is the magnetizing current density, and ν is the reluctivity, σ is the reluctivity.

In order to facilitate the analysis of the braking characteristics for this model, we neglected hysteresis losses and the highest harmonic and density of the displacement current. The inner and outer sides of the rotor have 12 salient poles. The thickness of the blades is 10 mm. Because of the periodic boundary conditions, only 1/12 of the whole region is analyzed, as shown in Fig. 6.

Both rotor and stator were manufactured in low-carbon steel (Steel 10). The B–H curve of Steel 10 is shown in Fig. 7. The air gaps between the salient poles and the stator are 0.7 and 1.2 mm. The diameter of the copper wire is 5 mm, and a special layer of high temperature resistant material surrounds the outer layer of the copper wire. The actual diameter is 8 mm. There are 108 turns of the copper wire, and the total current of the excitation coil is 70A. Fig. 8 a) shows the mesh of the model. The model was meshed using a hexahedral nodal element. There were both symmetric boundary and rotational periodic boundary conditions. The mesh size of the model is set to 3 mm. The skin depth parameter is set to 0.6 mm, and the number of grid layers is set to 3.The size of the elements in mesh directly influenced the calculation accuracy and time. The flux density distribution is shown in Fig. 8 b) during a transient state, when the speed was 1500 rpm and the exciting current 70 A. The most probable position for magnetic saturation is in the inner surface of the stator next to the salient pole of the rotor. The maximum flux density at that position is 1.65 T.

The purpose of obtaining the braking characteristics is to provide the

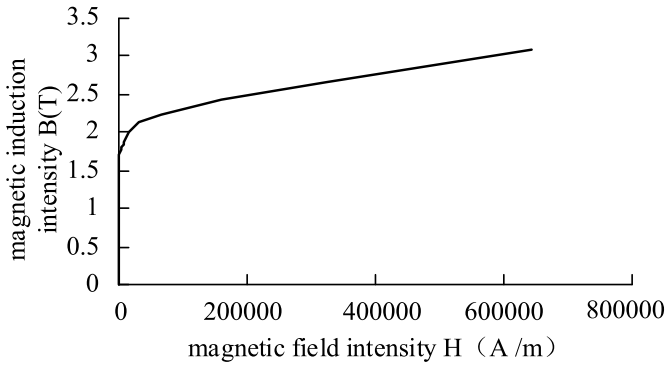
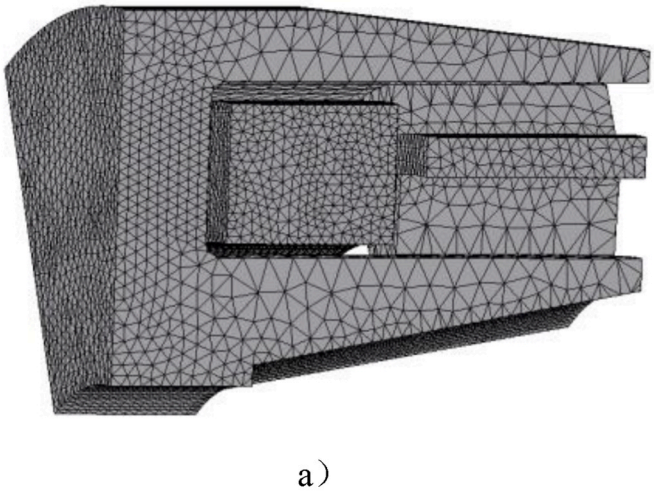
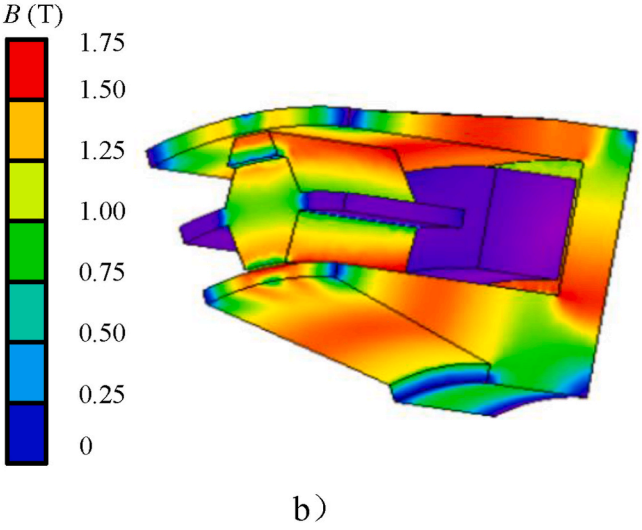


Fig. 7. The B–H curve of Steel 10.



a）



b）

Fig. 8. Mesh and the flux density distributions.

necessary data for an intelligent control system at various speeds and excitation currents. The braking characteristics curve is shown in Fig. 9. The braking torque increased rapidly with rising rotational speed and excitation current. When the speed is greater than 1000 rpm, the increments of the braking torque reduce.

Temperature rising tends to increase resistivity and decrease

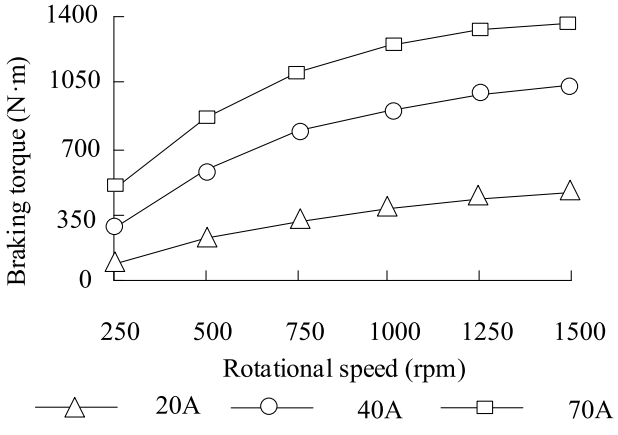


Fig. 9. Braking torque–current curve at various speeds.

permeability, which in turn diminishes the braking torque of the electromagnetic subsystem. Likewise, the rise of coolant temperature affects the braking torque of hydraulic subsystem. Fig. 10 presents the temperature distributions of the C-type stator at a speed of 1000 rpm, a braking torque of 1200 Nm, and different intervals of continuous braking. It can be seen that after 30 s, the temperature of the inner face of the stator was at 290 °C.

3.2. Flow field model analysis

In order to determine the main structural parameters of the flow field model, we use a mathematical description based on the one-dimensional flow beam theory (Wei et al., 2019, 2022). Both the transient internal flow field characteristics and the braking characteristics were analyzed.

The stator on both sides of the hydraulic part adopts a closed working chamber structure arranged symmetrically. The rotor blade adopts axisymmetric structure. The rotor and stator blades are forward curved axial straight blades, which can accelerate and turn the working fluid. Because the hydraulic part adopts axisymmetric design. In order to simplify the calculation difficulty, only one side of the working chamber flow field is selected to establish the model.

Based on the analysis results of the flow field in the impeller of the side channel pump with similar structure(Meakhail and Seung, 2005), the following assumptions are put forward:

1) The working fluid flows in from the inlet of the rotor impeller, and the rotor blade works on the working fluid to accelerate it.
2) The working fluid is divided on the blade, part of the working fluid flows out from the top of the impeller and turns, and the other part flows out from the side of the impeller due to axial deflection due to centrifugal action inside the impeller.
3) There is no obvious boundary between them, which can be equivalent to two streamline beams.

The flow field model of one circulatory circle is described in Fig. 11. The area labeled I is the inlet of the liquid, while area II is the outlet of the liquid. Due to the limited number of blades and the existence of axial eddy currents between the blades, the liquid flow direction at the inlet and outlet of the blades should be deflected by a certain angle relative to the tangential direction of the blade working face.

$$R_7 = 0.5(R_m + R_h) \tag{6}$$

$$R_s = 0.5(R_8 + R_m) \tag{7}$$

$$R_m = \sqrt{0.5\left(R_8^2 + R_h^2\right)} \tag{8}$$

where R_h is inner radius of one circulation circle, R_7 is inlet equivalent radius, R_m is the boundary equivalent radius of the blade inlet, R_8 is outer radius of one circulation circle, R_s is boundary equivalent radius of the blade outlet, and b is the axial width of the blade.

The braking torque of the flow field model is obtained from (MU et al., 2016; Ma et al., 2016):

$$M = \rho\eta\lambda n^2 D^5 \tag{9}$$

where ρ is the density of the working liquid, η is the charging ratio, λ is the torque coefficient, n is the rotational, and D is the effective diameter of the working chamber.

The full flow channel model was selected to analyze the non-periodic fluid flow. The stator and rotor of the flow field model are made of aluminum alloy. The distance between stator and rotor is 3 mm. We selected water as the working medium. The number of blades on rotor and stator are 24 and 26, respectively. In order to facilitate the meshing and calculation analysis, we developed a periodic flow channel analysis model. Fig. 12 shows the mesh of the flow field model in a static state.

The inlet and outlet boundary conditions were set according to the

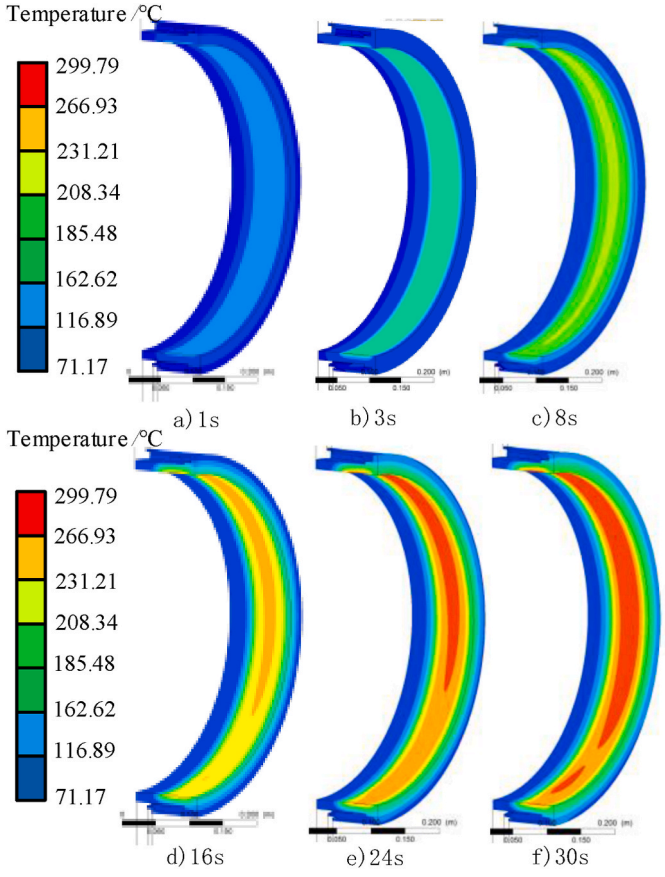


Fig. 10. Temperature distribution of the C-type stator.

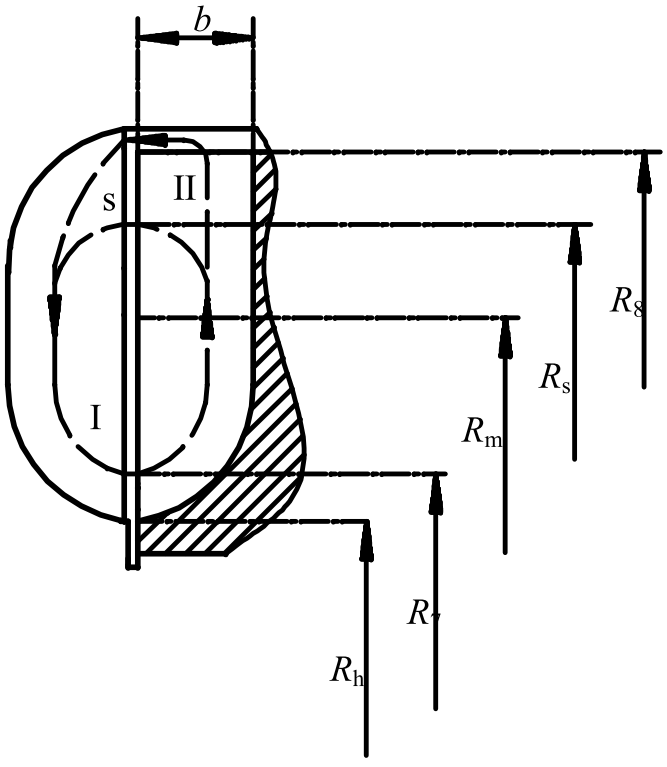


Fig. 11. Flow field model of one circulatory circle.

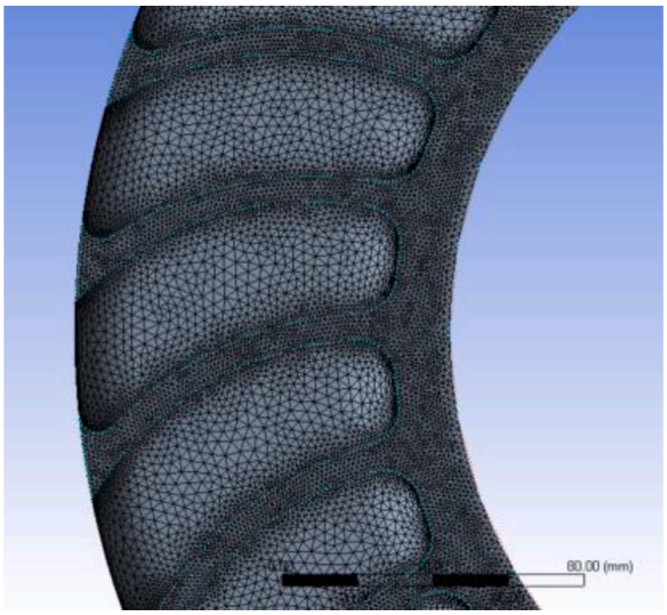


Fig. 12. Mesh of the flow field model.

actual working conditions. The outlet conditions were free outflow. In order to obtain the braking characteristics and parameters of the flow field model, we first derived the relationship between the rotor speed and the charging ratio. The purpose of obtaining the charging ratio was to provide data for an intelligent control system at various speeds.

Fig. 13 illustrates the velocity vector and pressure distribution of the fluid on the neutral surface of the circulatory circle. The center of the fluid is a low-pressure area. Since this negative pressure area is existing, the pressure increases along the radial direction. The maximum flow field pressure is at the bottom and top of the circulation circle, as shown in Fig. 13 (a). The working fluid accelerates from the top part to the bottom, as shown in Fig. 13 (b). The speed is higher at the center of the circulation circle. The velocity increases along the radial direction of the rotor. Fig. 14 shows the braking torque curve at various pressures.

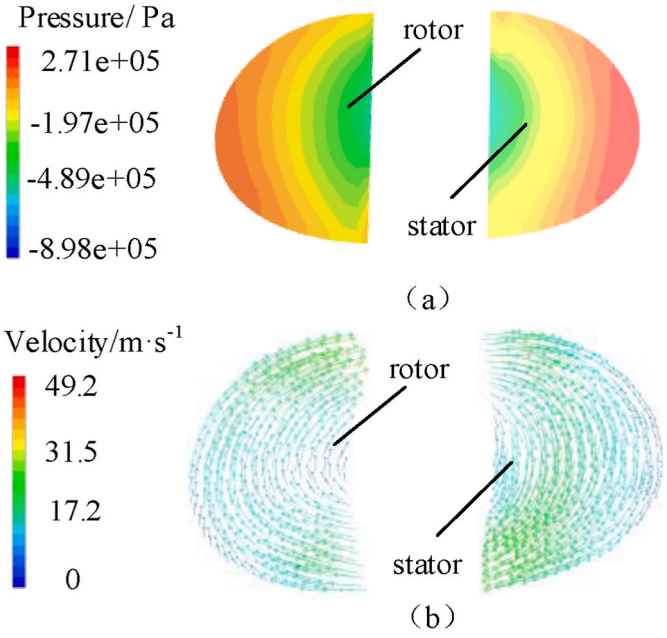


Fig. 13. Pressure and velocity distribution.
(a) Pressure. (b) Velocity vector field.

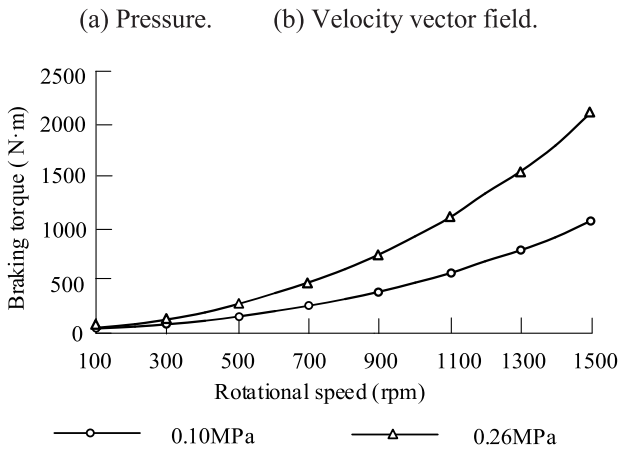


Fig. 14. Braking torque curve at various pressure.

3.3. Blend braking control strategy

The control system of the IEHR has the characteristics of multi disturbance, multi working conditions, strong coupling and nonlinearity. NIE Mengwen proposed a scheme of energy recovery type retarder installed in the hub on both sides of the front axle of semi-trailer, using particle swarm optimization algorithm for braking torque distribution (Mengwen et al., 2021). A number of the recent literatures have been published on a regenerative braking controller trained with an Extended Kalman Filter (EduardoSanchez Edgar et al., 2021), and fuzzy logic model for regenerative braking for modeling Electric Vehicles' regenerative braking systems (Maia et al., 2021). To improve the IEHR utilization efficiency, we design a braking force distribution strategy based on a fuzzy control system.

The control objectives of blend braking control strategy include the braking strength, the braking torque of the electromagnetic part and the braking torque of the hydraulic part, which are represented by braking strength *z*, excitation current *I* and system pressure *P*. The controller makes comprehensive decisions based on z, *I*, and *P*, and selects the braking strategy mode in a braking cycle. According to the braking strategy mode and the braking demand, the braking force is distributed to friction braking, electromagnetic braking and hydraulic braking. Then, return to the beginning and judge the next cycle. The diagram of the blend braking control strategy is shown in Fig. 15.

The control system adopts the special embedded chip for automobile as the core, and uses pulse width modulation and power module to

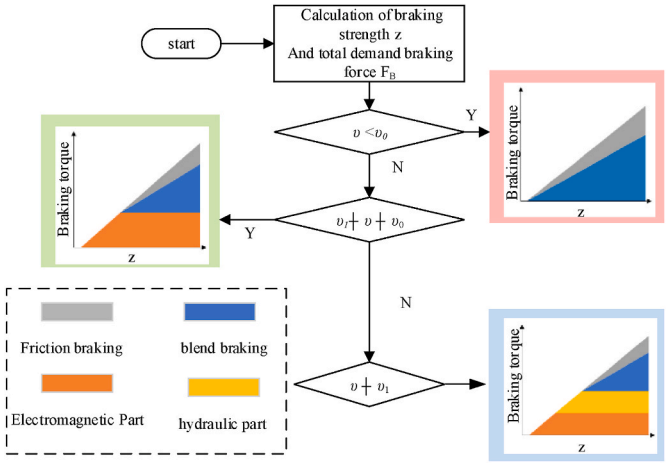


Fig. 15. Diagram of the blend braking control strategy.

drive. A closed-loop control system is formed by receiving and processing signals from vehicle speed, outer rotor temperature, cruise control, excitation current and can bus. In order to match the retarder with the vehicle braking demand, the required braking strength z is calculated according to the braking pedal travel. Different braking strategies shall be adopted according to different speeds. When the speed is lower than ν_0 km/h, the braking of electromagnetic part can be adopted. When the speed is between ν_0 km/h and ν_1 km/h, the hydraulic braking and electromagnetic braking shall be adopted at the same time. When the speed is exceeds ν_1 km/h, the hydraulic braking and electromagnetic braking and friction braking shall be adopted. In case of emergency braking, the braking strategy is the same as when the speed is exceeds ν_1 km/h.

With the continuous changes in vehicle speed and braking intensity, the braking strategy mode frequently switches during the driving process. The function of the fuzzy logic control strategy for IEHR is to distribute the braking torque among friction braking, hydraulic braking and electromagnetic braking. In order to simplify the fuzzy logic, three fuzzy language variables, large, medium and small, are selected to describe the input and output of the controller, such as friction braking f_1, hydraulic braking k_1, electromagnetic braking k_2, and braking intensity z. According to the theoretical analysis and simulation results, the membership functions of variable is shown in Fig. 16.

Fuzzy rules are summarized according to simulation results. The fuzzy controller is designed by using Matlab. The fuzzy inference rule adopts Mandani method. Fuzzy logic toolbox can generate fuzzy reasoning surface according to fuzzy rules. The rule of fuzzy control is shown in Fig. 17.

The control system of the IEHR has the characteristics of multi disturbance, multi working conditions, strong coupling and nonlinearity. The effectiveness of the IEHR used on heavy vehicles is evaluated by braking distance and braking deceleration. When heavy vehicles go downhill, the target speed is set to 30 km/h and cruise control is achieved using the IEHR, as shown in Fig. 18.

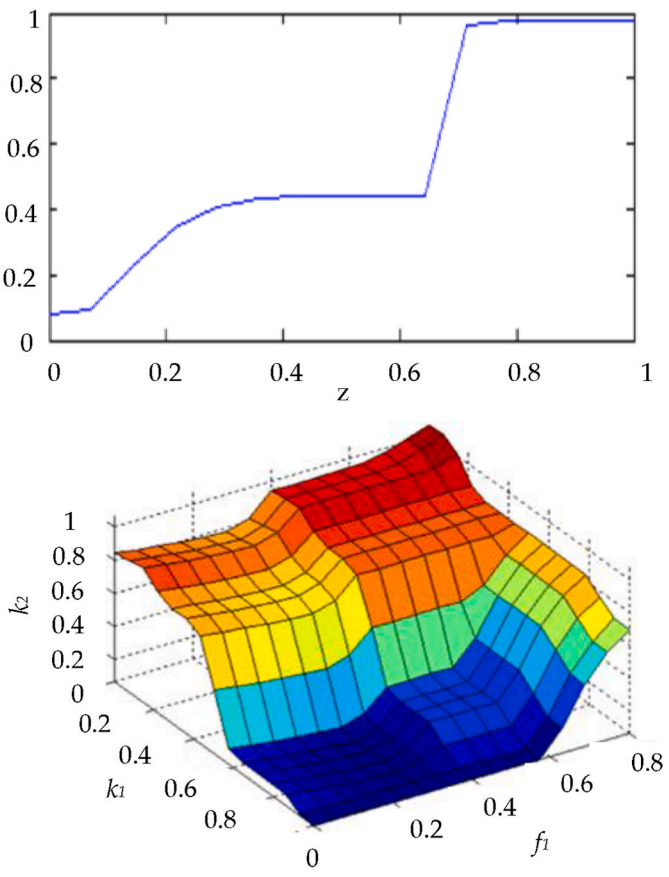


Fig. 17. The rule of fuzzy control.

4. Experiment

Based on the above-described analysis, we developed a prototype of the IEHR. In order to test the IEHR performance, the braking characteristic was measured using a test bench. The test bench is made up of a 400 kW DC motor, a speed regulating device, the radiator, the water tank and the torque indicator. The braking torque is obtained by a torque indicator installed on the output shaft of the DC motor. The test data were collected by a computerized control system. The test bench is shown in Fig. 19. The braking torque of hydraulic part is determined by adjusting the system pressure. The system pressure is through adjusting the pressure of the water tank. The set value of system pressure is 0.1 MPa and 0.26 MPa through the above analysis. The radiator is to reduce the system temperature. The excitation current is provided for the battery, and the regulation range is 0~70A.

The main geometrical parameters of the analyzed model are shown in Table 1.

4.1. Braking characteristics of the prototype

We first obtained no-load characteristics to reduce the additional load when the IEHR was not operating. The no-load characteristics affects the economy of the IEHR. The total loss of the system including friction and no-load losses are shown in Fig. 20. The total loss increases with higher speeds. When the speed is 1500 rpm, the power loss is 1.4 kW. This meets the design requirement of having the no-load power losses below 2 kW.

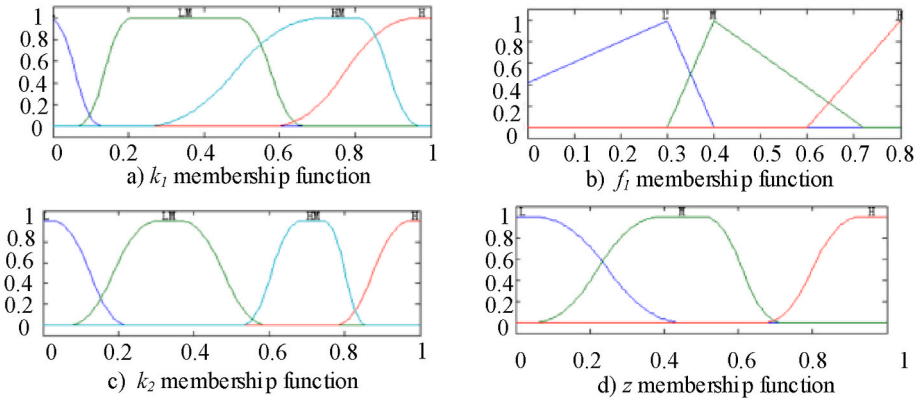


Fig. 16. Membership Functions of variable.

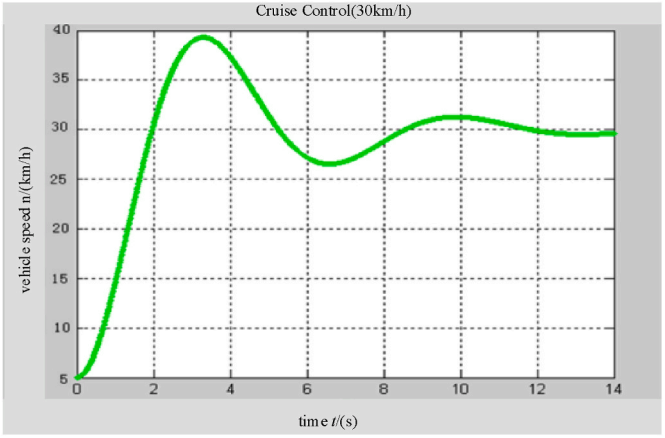


Fig. 18. Simulation results of constant speed fuzzy control.

During the experiment, the braking torque of the hydraulic part was controlled by adjusting the pressure in the working chamber. We obtained the curves of the braking torque under a pressure of 0.1 MPa and 0.26 MPa at various speeds, as shown in Fig. 21. When testing the braking characteristics of the hydraulic subsystem, the working fluid is provided by the external water tank. We used the braking characteristics to measure the braking ability of the IEHR and to facilitate grading control. It can be seen that the braking torque is small when the speed is less than 500 rpm. On the other hand, when the speed is higher than 500 rpm, the braking torque increases rapidly. However, when the speed is higher than 1,800 rpm, the braking torque decreases again. The temperature rise is an important cause of the braking torque decrease. Other reasons that affect the braking torque are the physical properties of the working fluid, such as viscosity and density.

The braking torque of the electromagnetic and hydraulic subsystems were obtained for different speeds, providing a theoretical basis for the adjustment methods. When measuring the braking characteristics of the

Table 1
Parameters of the IEHR.

Symbol	Value	Symbol	Value
R_1 (mm)	125	R_8 (mm)	229
R_2 (mm)	162	h_1 (mm)	28
R_3 (mm)	163.2	h_2 (mm)	22
R_4 (mm)	222.2	H_1 (mm)	153
R_5 (mm)	233	H_2 (mm)	71
R_6 (mm)	255	H_3 (mm)	23
R_7 (mm)	140	H_4 (mm)	38

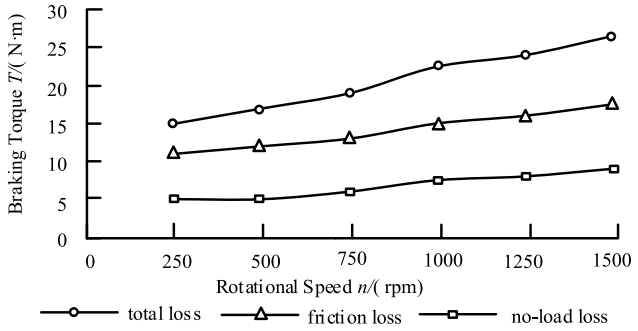
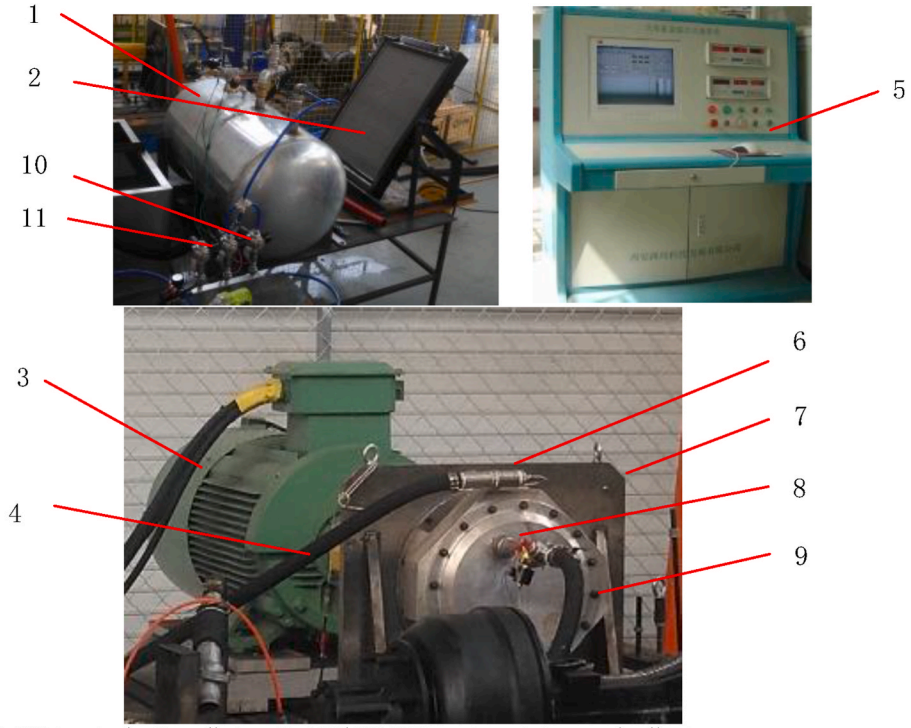


Fig. 20. No-load characteristics curves of the IEHR.



1-Water tank; 2-radiator; 3-400kW DC motor; 4-torque indicator;
5-computer-centralized control system; 6-water pipe;
7- fixation apparatus; 8-solenoid valve; 9-the IEHR;
10-0.1MPa solenoid valve; 11-0.26MPa solenoid valve

Fig. 19. Test bench of the IEHR.

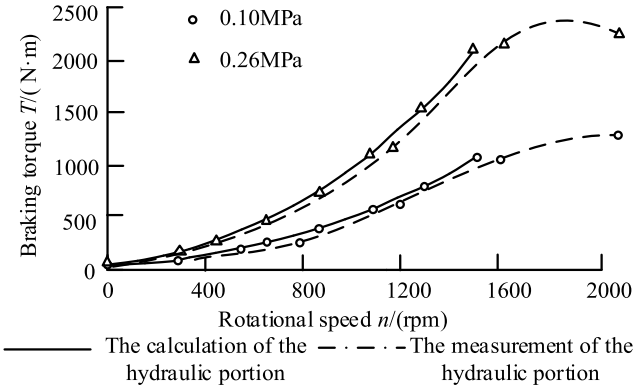


Fig. 21. Braking torque curves at various pressures.

electromagnetic subsystem, the excitation current is provided by the vehicle battery. We obtained the curves of the braking torque at various current, as shown in Fig. 22. The difference between the results and the design requirements was then 10%. Considering the magnetic saturation and the influence of the rise of the temperature, the relationship between the braking power and the rotational speed was found to be nonlinear. There are many factors affecting the correlation of the results, including the flux leakage, the magnetic saturation, the rise of the temperature, inaccurate machining tolerances of the components, and other factors. The most important factor is the influence of the temperature. When the temperature of the inner stator reached 200 °C, the magnetic resistance rate of the stator material increased.

The experimental values of the braking torque at various speeds, a pressure of 0.26 MPa and an excitation current of 70 A are shown in Fig. 23. For comparison, we have overplot the theoretically calculated braking torques of the electromagnetic and hydraulic subsystems. Because of the increasing temperature of the stator, the saturated magnetic density decreases. The braking torque follows a downward trend as the speed increases. It can be seen from the diagram that the theoretical and experimental values of the electromagnetic braking torque are basically the same. Comparing with the test results, the calculation error of the flow field finite element model considering the number of grids, calculation time steps and temperature factors is reduced to 5.8%–6.4%. The braking torque peaks at 1283 Nm for a speed of 1500 rpm and an exciting current of 70 A.

The braking torque of the hydraulic subsystem is relatively small at low speeds, only 113 Nm at 300 rpm. On the other hand, for a speed of

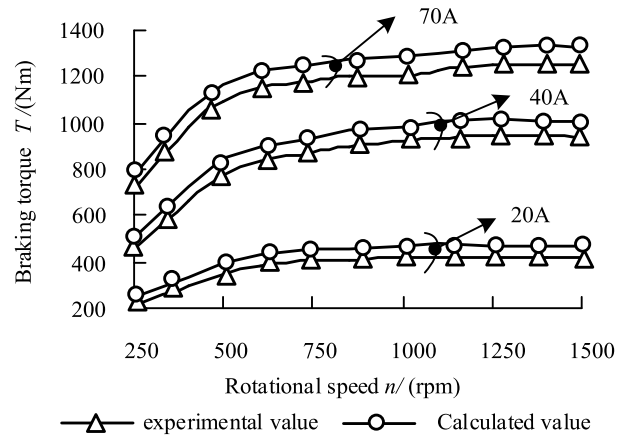


Fig. 22. Comparison of measured and calculated braking torque at various speeds.

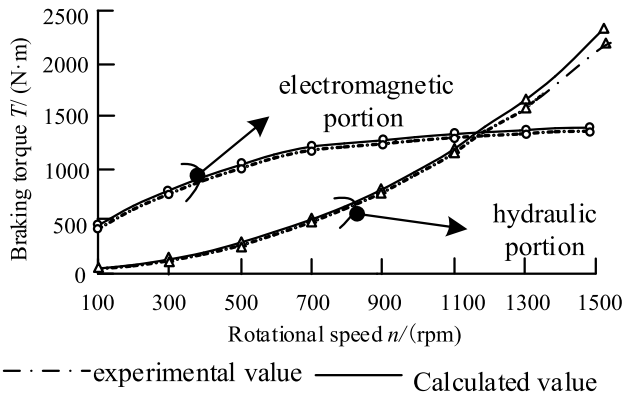


Fig. 23. Comparison of experimental and calculated braking torques of both subsystems.

1500 rpm, the braking torque reached 2128 Nm. The change in the trend of the braking torque is basically consistent with that of traditional HRs, under similar conditions. The braking torque of the electromagnetic subsystem reached 1050 Nm at a speed of 500 rpm. Showing that electromagnetic braking is more suitable in a low speed regime of the vehicle, while hydraulic braking performs better at high speed. Therefore, combined hydraulic and electromagnetic braking is suitable for every speed stage of the vehicle.

Fig. 24 shows the measured and theoretical breaking torque characteristics at various speeds. There are many non-modeled factors affecting the agreement between calculated and measured torques, including magnetic saturation, flux leakage, the reduction of the working chamber charging ratio as the speed increases, tolerances of the manufactured components, etc. The maximum difference between the experimental results and the design requirements is then 10%. At a speed of 1500 rpm, a pressure of 0.26 MPa and an exciting current of 70 A, the braking torque reached 3620 N m.

In theoretical analysis, the braking torque of IEHR is the sum of the hydraulic braking torque T_L and the eddy current braking torque T_E. In the actual test, the internal liquid temperature of the hydraulic retarder rises when it works. As the heat dissipation medium of the eddy current retarder, the liquid affects the heat dissipation effect. As a result, the braking torque of IEHR decreases slightly when the IEHR is running.

$$T = T_L + T_E \tag{10}$$

The comparison curve between the IEHR and traditional hydraulic retarder is shown in Fig. 25. The braking torque of the traditional HRs is very small at low speed. The higher the speed, the greater the torque.

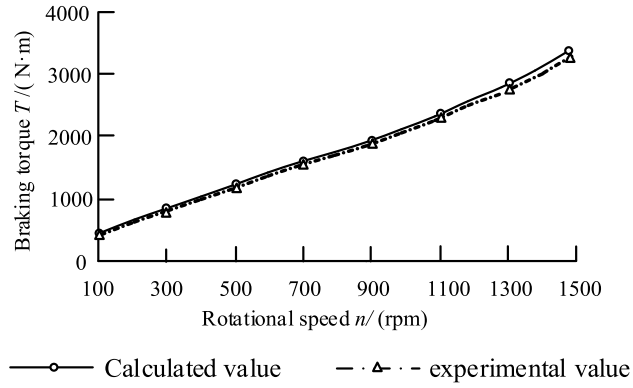


Fig. 24. Comparison of the braking characteristics obtained by both experimental and calculated.

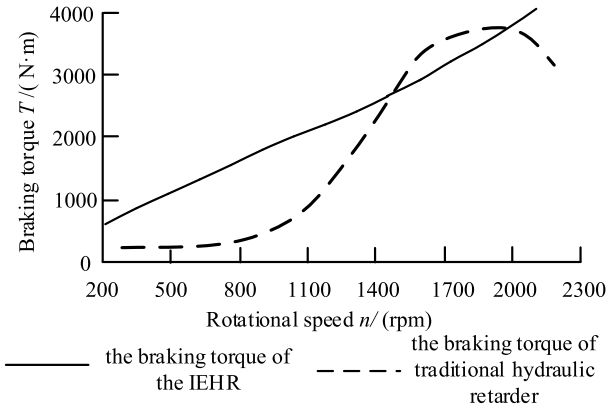


Fig. 25. The comparison 2curve between the IEHR and traditional hydraulic retarder.

However, the braking torque starts to decrease with the increase of the speed due to the increase of internal fluid temperature. The control strategy of the traditional HRs is to reduce torque when the temperature is high. That is, the eddy current retarder's braking torque is great at the low speed, and its control is simple; at the high speed, the hydraulic retarder's braking torque is large. Based on this, the IEHR is designed by increasing a small amount of manufacturing cost to obtain the advantages of high braking torque at full speed and simple control.

4.2. Road test

The purpose of road test is to explore the braking of the IEHR under realistic conditions. The braking torques of hydraulic and electromagnetic subsystems are reasonably distributed to facilitate hierarchical braking and fulfill the vehicle braking requirements. The manufactured IEHR was assembled on a test heavy-duty vehicle (Centre trailer of car carrier). The main parameters of the heavy-duty vehicle are shown in Table 2. The total mass of the vehicle is 49,000 kg. The current in the excitation coil is 70A, and the pressure of the working chamber 0.26 MPa. The IEHR was installed in front of the vehicle axle. The controller is installed on the beam. A Hand-switch and a luminous indicator were installed in cab, see Fig. 26. There are differences in retarder evaluation standards of different countries. Taking the national standard of China as an example, the test method of braking system of automobile retarder is specified. The standard stipulates that the deceleration shall be at least 0.6 m/s^2 when the vehicle speed is reduced from 50 km/h to 10 km/h.

Fig. 27 a) presents the data acquisition system. The collected data include vehicle speed and acceleration, and water temperature at the outlet of the IEHR. Vertical gyroscope (VG700AB) is used to accurately measure the road slope in this experiment, which is specially designed for vehicle dynamic test, and can provide high-precision vertical gyroscope and inertial measurement unit output. The temperature sensor was installed at the retarder outlet. Fig. 27 b) presents the interface of the vehicle data acquisition system.

The effectiveness of the IEHR used on heavy vehicles is evaluated by braking distance and braking deceleration. The braking deceleration was obtained using the IEHR brake at both low speed and high speed. The braking time and deceleration when the vehicle speed is reduced from 50 km/h to 10 km/h, are shown in Table 3. The braking time and deceleration are 14.6 s and 0.75 m/s^2 respectively. On the other hand, these quantities are, 17.1 s and 0.65 m/s^2.for speed reduction from 70 km/h to 30 km/h.

Table 2
Main parameters of the tested vehicle.

parameter	*Value*
Total mass (kg)	49,000
Maximum IEHR torque (Nm)	3500
Current of excitation coil (A)	70
Pressure of the system (MPa)	0.26

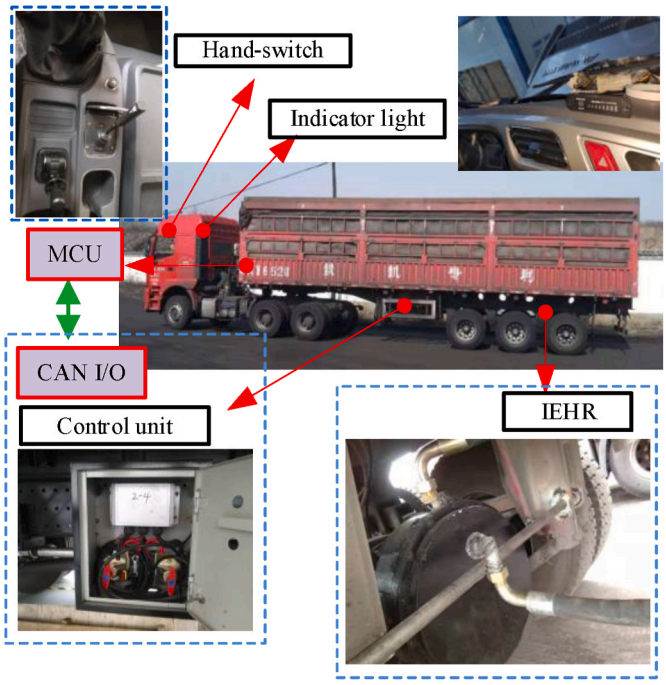


Fig. 26. Parameters of the IEHR assembled on a test, heavy-duty vehicle.

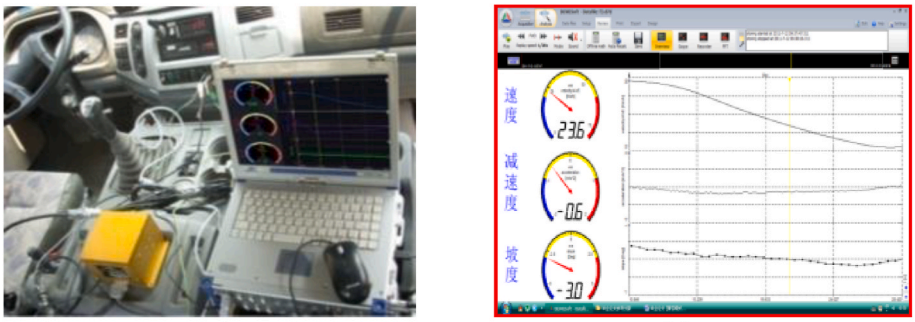
When the vehicle is going downhill, the IEHR is often used to assist braking by consuming kinetic and potential energy. The industry standard requires the braking deceleration of the vehicle to be greater than 0.6 m/s^2 when just relying on the retarder. The braking deceleration meets the national standard and the vehicle braking demand. The IEHR converts the kinetic energy into heat, which is taken away by the cooling system. Compared to the conventional HRs, our proposed IEHR has a high energy density below 30 km/h, providing better braking performance. The road test results show that the IEHR has good braking performance at both low speed and high speed.

5. Conclusions

This paper proposes an Integrated Electromagnetic and Hydraulic Retarder suitable to be used in heavy-duty vehicles. We perform theoretical, numerical and experimental studies. The produced prototype is tested both in a test-bench and the road. The measured braking characteristics of the IEHR are analyzed and compared to the theoretical results finding good agreement. The braking characteristics of the electromagnetic and hydraulic subsystems are analyzed independently, further supporting our theoretical method for the IEHR controlling at different speeds.

The manufactured IEHR was assembled on a tested heavy-duty vehicle. The braking time and deceleration when the vehicle speed is reduced from 50 to 10 km/h, are 14.6 s and 0.75 m/s^2 respectively, and is more suitable for heavy vehicles. The braking characteristics of the ECRs and the HRs are complementary, thus IEHRs have the advantages of both of them. Compared to conventional HRs, our IEHR has a high energy density below 30 km/h. This excellent performance makes it suitable for heavy-duty vehicles.

The IEHR converts the kinetic energy of the vehicle into heat energy. In the future research, we will focus on optimizing the control strategies, energy recovery and its utilization for fuel cells and/or cab heating. And the IEHR will be installed on the electric bus. The retarder provided the



a) b)

Fig. 27. Data acquisition system for the IEHR
a) Data test b) The interface of data acquisition system.

Table 3
Test data of the heavy-duty Vehicle.

parameter		Value I	Value II
Speed range(km/h)		50–10	70–30
Braking time (s)		14.6	17.1
Braking deceleration (m/s^2)		0.76	0.65
temperature (°C)	inlet	20	21
	Outlet	81~115	92~120

generated heat energy to the cabin through the heat exchanger. It can reduce battery consumption and improves the vehicle mileage.

CRediT authorship contribution statement

Kai Zhang: IEHR Structural design, Methodology, creation of models, Analytical method, numerical simulation, control system design, Road test. **Huichao Shang:** Structural design. **Jing Xu:** numerical simulation, optimization. **Jigao Niu:** control system design, Programming, Software. **Yonggao Yue:** Road test.

Declaration of competing interest

The authors declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Testing and Performance Analysis of an Integrated Electromagnetic and Hydraulic Retarder for Heavy-duty Vehicles”.

Data availability

No data was used for the research described in the article.

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